

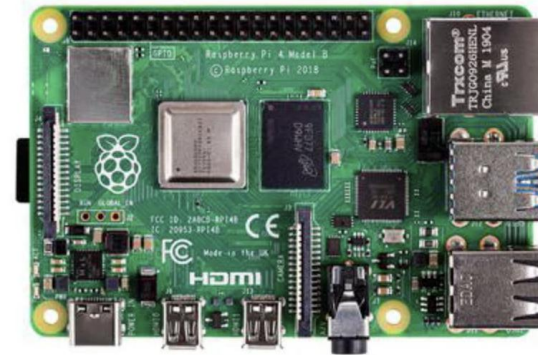
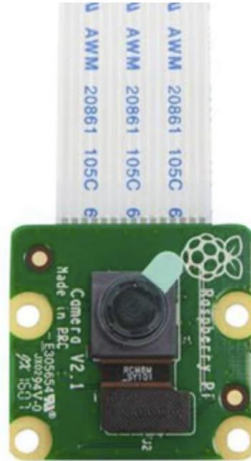
SAN FRANCISCO STATE UNIVERSITY SCHOOL OF ENGINEERING

On-Device Money Detector and Calculator

Presenter: Milton Tinoco

Design Project

- ▶ Develop a machine learning model for embedded systems (e.g., Raspberry pi, smartphones, Arduino Nano 33).
- ▶ Model Enable recognition of coins and cash denominations.
- ▶ Calculate totals accurately.

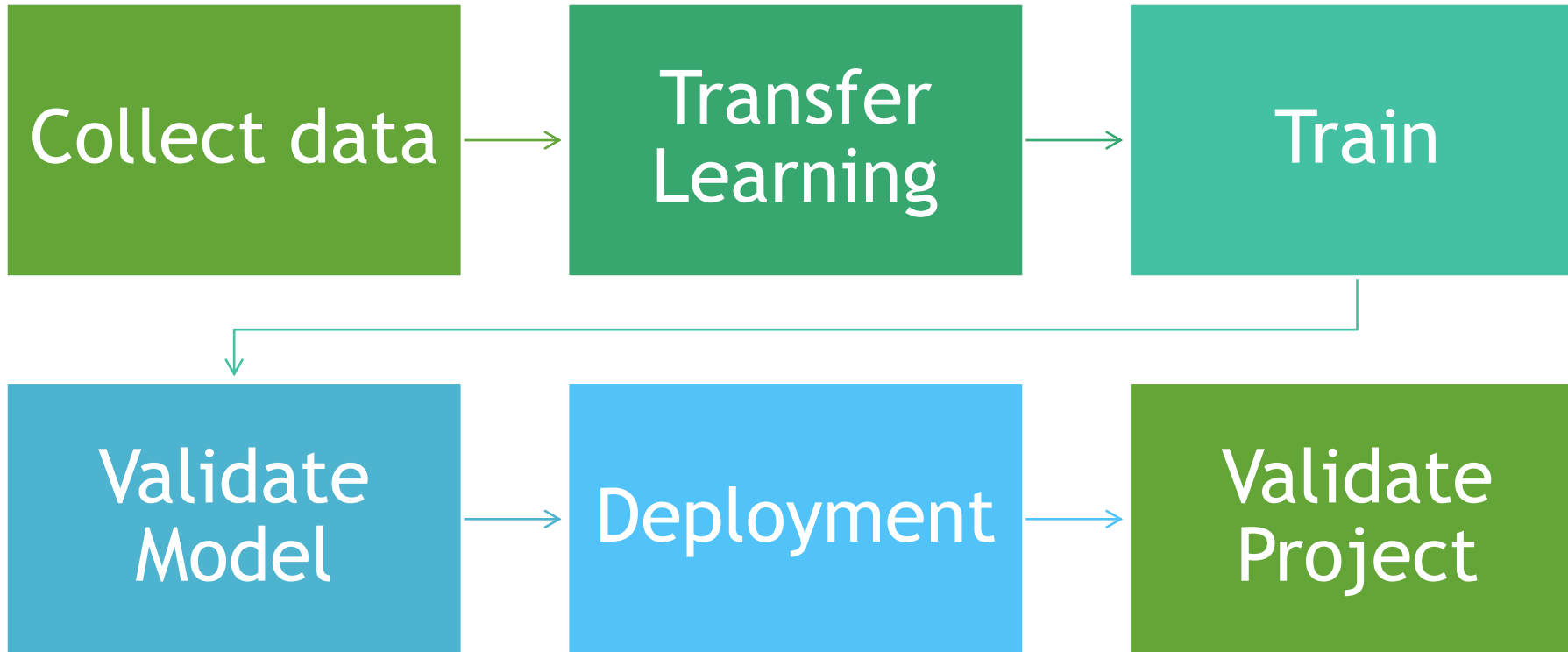


Motivation

- ▶ Enhance currency recognition accuracy, reduce costs and errors in point-of-sale transactions
- ▶ expand its application to industries like retail, banking, and vending, thereby improving operational efficiency and user experiences
- ▶ contributing to societal well-being and technological advancement.



Plan





Collect Data

800 pictures diversity of US Bills and Coins

Mistake:

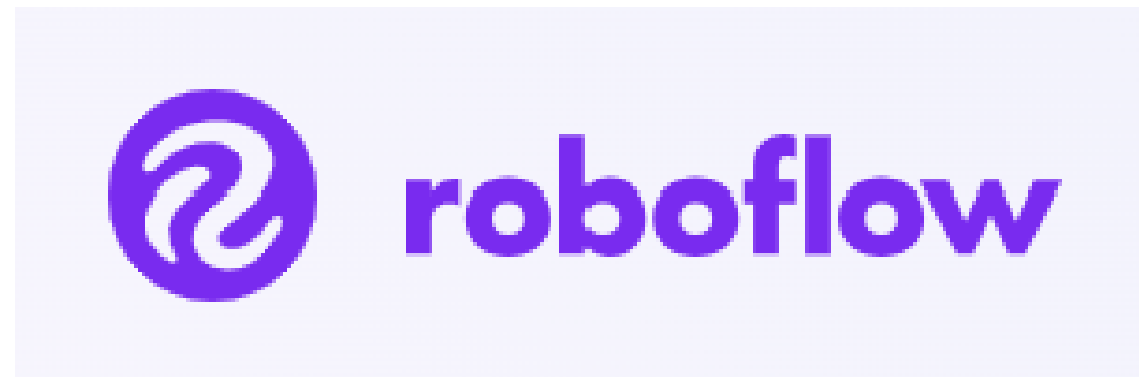
Too many
pictures

Make
Sense AI

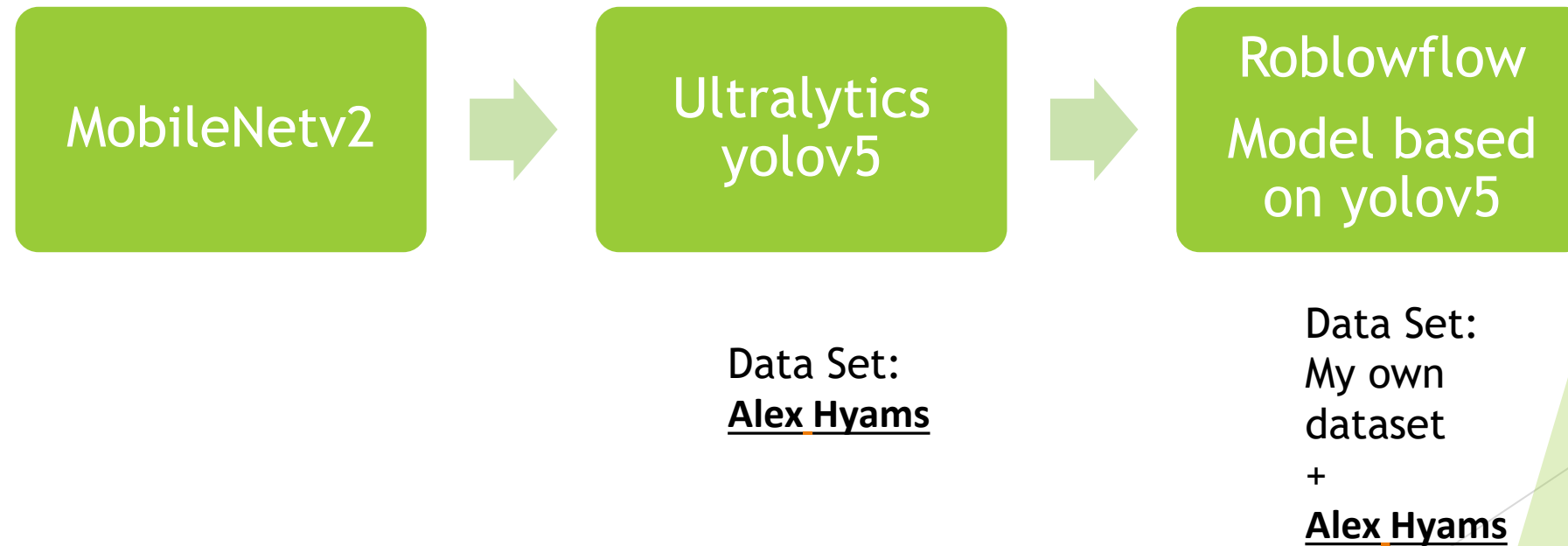
CVAT



Solution:



Transferring Learning



Training:

Roboflow Train

Automatic training on Roboflow's GPU Cluster for a [deployable](#) » model within a few hours.

AVAILABLE TRAINING CREDITS: 1 [Upgrade Plan](#) »

Model Type: Roboflow 3.0 Object Detection (Fast)

RECOMMENDED

🚩 Train from Previous Checkpoint

Start from one of your previous training runs to speed up training and improve accuracy. This option is best if you already successfully trained a model on this project.

NOTE: Only models with the same model type as above are available as checkpoints.

Select Model

MoneyMiltonProject

Select Model Version

v4 - 2024-05-19 11:36am (onfaceup-onefacedown-quarterup-, 81.3% mAP)

☆ Train from Public Checkpoint

Use a pre-trained benchmark model or a starred Universe project to imbue your model with prior knowledge, reduce training time, and improve performance.

🎲 Train from Random Initialization

Not recommended; almost always produces worse results.

Cancel

Start Training

```
v5 > main.py > ...
import torch
from pathlib import Path

def train_yolo(image_size, batch_size, num_epochs, path_to_config_yaml, path_to_model_config_yaml, optional_pretrained_weights):
    # Training command
    train_command = f"python train.py --img {image_size} --batch {batch_size} --epochs {num_epochs} --data {path_to_config_yaml} --cfg {path_to_model_config_yaml}"

    # Run training
    subprocess.run(train_command, shell=True)

    # Path to the saved trained model
    trained_model_path = Path("runs/train/exp/weights/best.pt")

    return

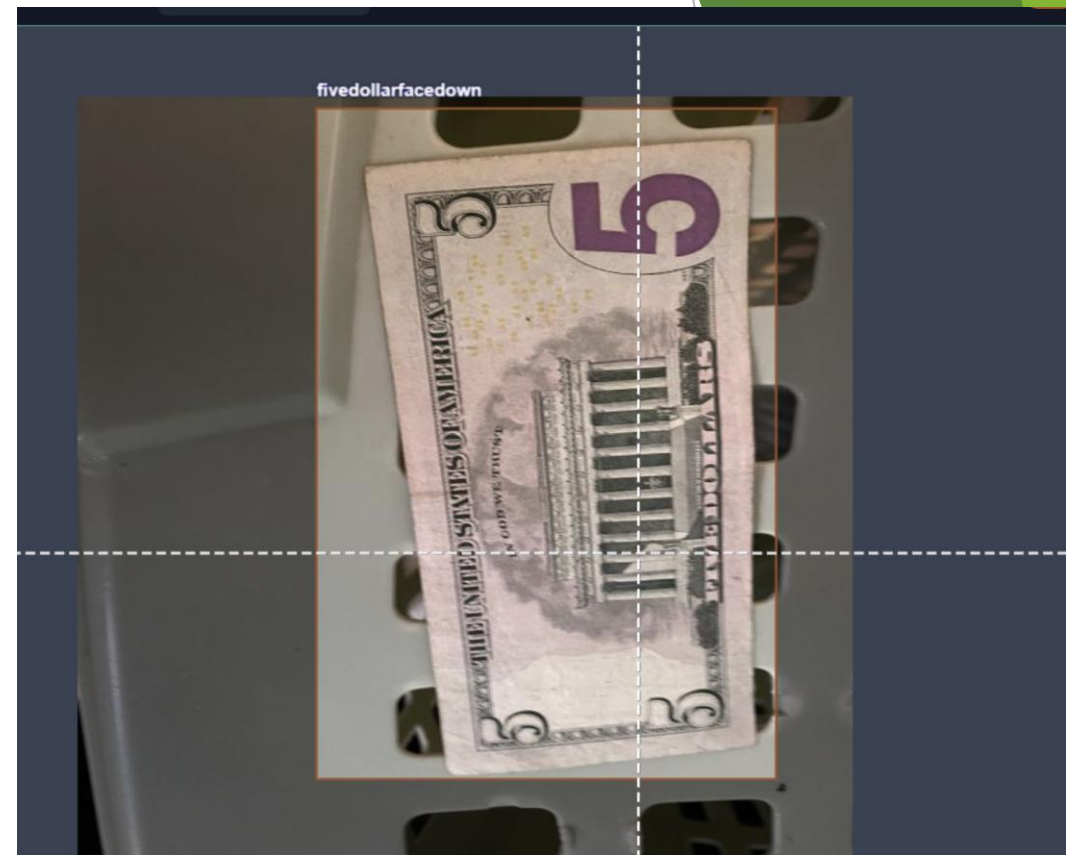
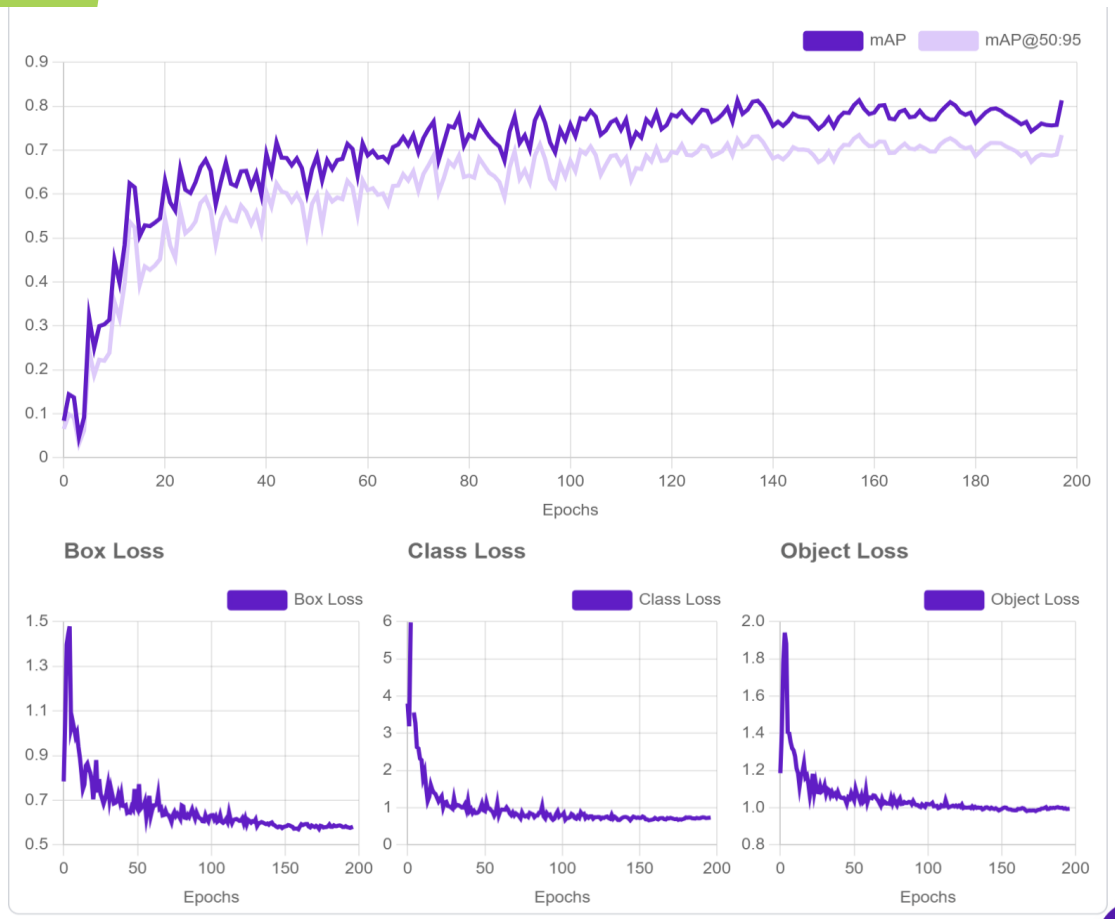
def export_model(model_path, export_format):
    # Export command
    export_command = f"python export.py --weights {model_path} --include {export_format}"

    # Run export
    subprocess.run(export_command, shell=True)

# Example usage:
image_size = 300
batch_size = 16
num_epochs = 200
path_to_config_yaml = r"C:\Users\milto\OneDrive\Desktop\Version2\MoneyMiltonProject-1\data.yaml"
path_to_model_config_yaml = "models/yolov5s.yaml"
optional_pretrained_weights = r"C:\Users\milto\OneDrive\Desktop\Version2\yolov5\yolov5s.pt"

# Train the model
trained_model_path = train_yolo(image_size, batch_size, num_epochs, path_to_config_yaml, path_to_model_config_yaml, optional_pretrained_weights)

# Export the trained model to OpenVINO format
export_model(trained_model_path, "openvino")
```

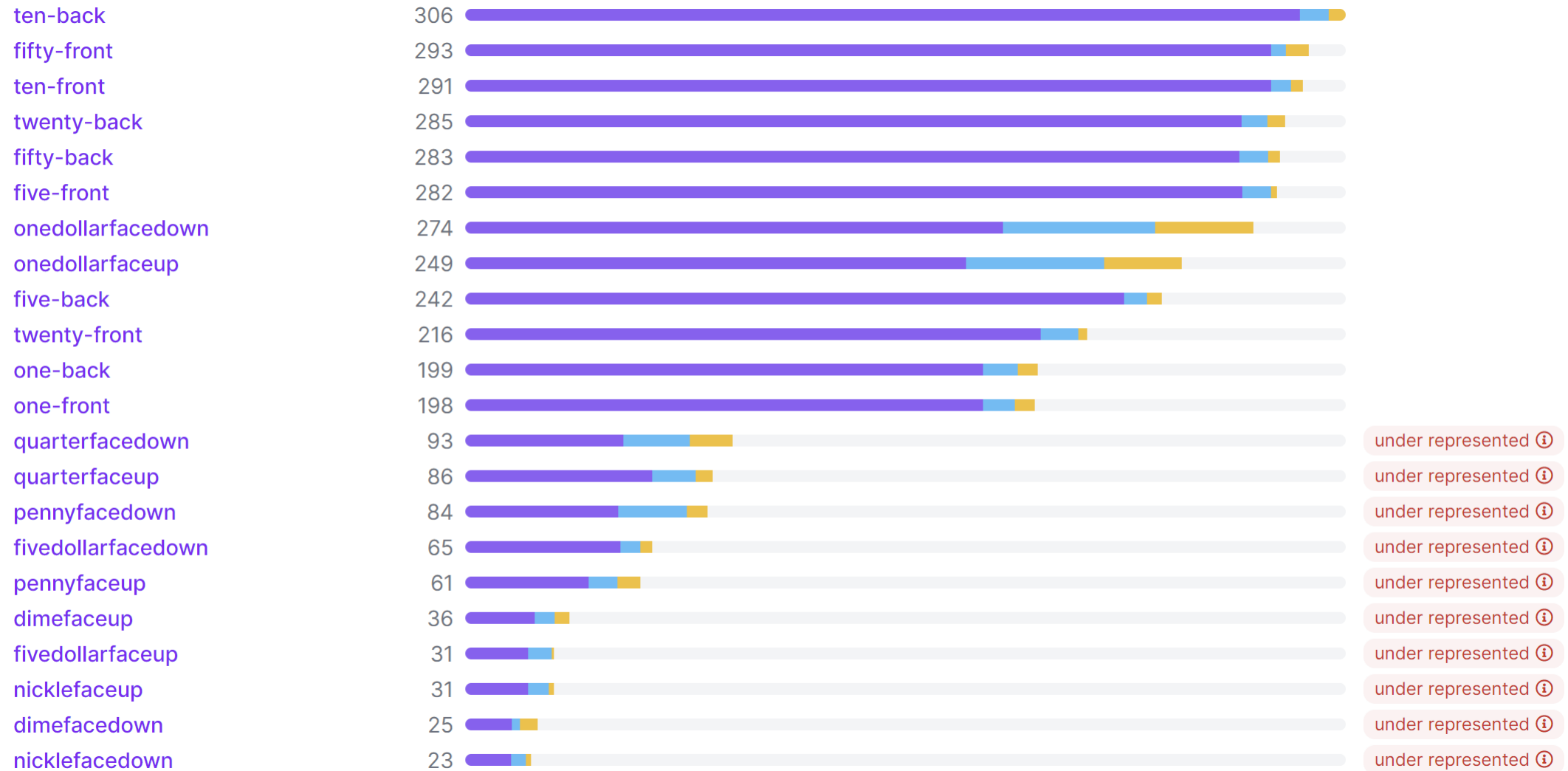
Model Validation

Roboflow

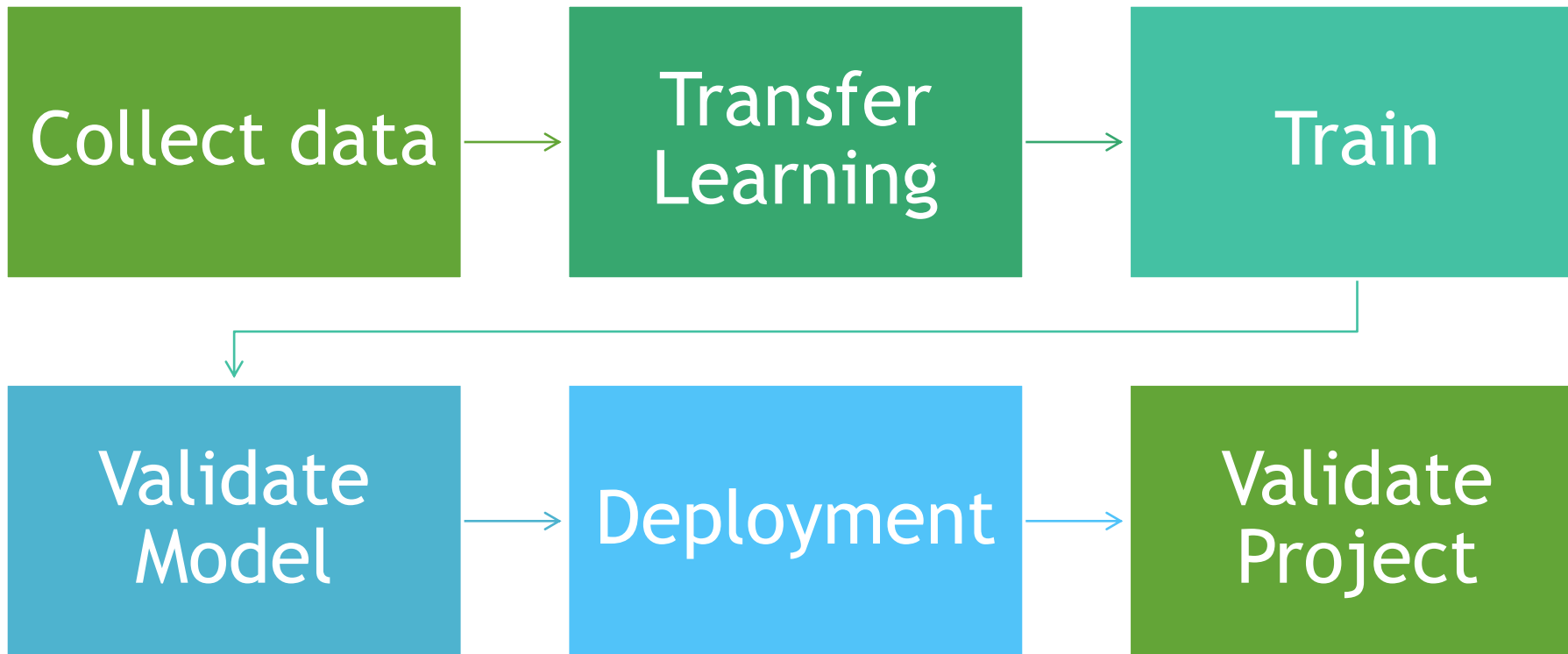


Model Validation Roboflow

Data Set:



Plan





```
YOLOv5s summary: 213 layers, 7225885 parameters, 0 gradients, 16.4 GFLOPs  
1/1: tcp://127.0.0.1:8888... Success (inf frames 1280x960 at 1.00 FPS)
```

Deployment
Raspberry pi using pi camera

```

(Homework6) PS C:\Users\milto\OneDrive\Desktop\Version2\yolov5> python export.py --weights C:\Users\milto\OneDrive\Desktop\Version2\yolov5\runs\train\exp7\
weights\best.pt --include torchscript
export: data=C:\Users\milto\OneDrive\Desktop\Version2\yolov5\data\coco128.yaml, weights=['C:\\Users\\milto\\OneDrive\\Desktop\\Version2\\yolov5\\runs\\trai
n\\exp7\\weights\\best.pt'], imgsz=[640, 640], batch_size=1, device=cpu, half=False, inplace=False, keras=False, optimize=False, int8=False, per_tensor=False,
dynamic=False, simplify=False, opset=17, verbose=False, workspace=4, nms=False, agnostic_nms=False, topk_per_class=100, topk_all=100, iou_thres=0.45, c
onf_thres=0.25, include=['torchscript']
YOLOv5 v7.0-312-g1bcd17ee Python-3.9.19 torch-2.3.0+cpu CPU

Fusing layers...
YOLOv5s summary: 157 layers, 7037095 parameters, 0 gradients, 15.8 GFLOPs

PyTorch: starting from C:\Users\milto\OneDrive\Desktop\Version2\yolov5\runs\train\exp7\weights\best.pt with output shape (1, 25200, 15) (54.3 MB)

TorchScript: starting export with torch 2.3.0+cpu...
TorchScript: export success 1.6s, saved as C:\Users\milto\OneDrive\Desktop\Version2\yolov5\runs\train\exp7\weights\best.torchscript (27.3 MB)

Export complete (2.2s)
Results saved to C:\Users\milto\OneDrive\Desktop\Version2\yolov5\runs\train\exp7\weights
Detect:      python detect.py --weights C:\Users\milto\OneDrive\Desktop\Version2\yolov5\runs\train\exp7\weights\best.torchscript
Validate:    python val.py --weights C:\Users\milto\OneDrive\Desktop\Version2\yolov5\runs\train\exp7\weights\best.torchscript
PyTorch Hub: model = torch.hub.load('ultralytics/yolov5', 'custom', 'C:\Users\milto\OneDrive\Desktop\Version2\yolov5\runs\train\exp7\weights\best.torch
script')
Visualize:   https://netron.app
(Homework6) PS C:\Users\milto\OneDrive\Desktop\Version2\yolov5>

```

Export the model:

```
^^^^^^^^^^^^^^^^^^^^
File "/usr/lib/python3.11/pathlib.py", line 874, in __new__
    raise NotImplementedError("cannot instantiate %r on your system"
NotImplementedError: cannot instantiate 'WindowsPath' on your system
(859env) milto@milton:~/yolov5 $
```

Deployment

Format	`export.py --include`	Model
---	---	---
PyTorch	-	yolov5s.pt
TorchScript	`torchscript`	yolov5s.torchscript
ONNX	`onnx`	yolov5s.onnx
OpenVINO	`openvino`	yolov5s_openvino_model/
TensorRT	`engine`	yolov5s.engine
CoreML	`coreml`	yolov5s.mlmodel
TensorFlow SavedModel	`saved_model`	yolov5s_saved_model/
TensorFlow GraphDef	`pb`	yolov5s.pb
TensorFlow Lite	`tflite`	yolov5s.tflite
TensorFlow Edge TPU	`edgetpu`	yolov5s_edgetpu.tflite
TensorFlow.js	`tfjs`	yolov5s_web_model/
PaddlePaddle	`paddle`	yolov5s_paddle_model/


```
^^^^^^^^^^^^^^^^^^^^
File "/usr/lib/python3.11/pathlib.py", line 874, in __new__
    raise NotImplementedError("cannot instantiate %r on your system"
NotImplementedError: cannot instantiate 'WindowsPath' on your system
(859env) milto@milton:~/yolov5 $
```

Deployment



Let's Start Over

- ▶ Start Using Roboflow
- ▶ Trained
- ▶ Got great results
- ▶ Ready to deploy
- ▶ And then

```
raise NotImplementedError('cannot instantiate {} on your system'.format(model_type))
NotImplementedError: cannot instantiate 'WindowsPath' on your system
(859env) milto@milton:~/yolov5 $ raise DatasetLoadError(f"Model type not supported: {model_type}")
inference.core.exceptions.DatasetLoadError: Model type not supported: ('object-detection', 'yolo_nas')
INFO: 172.17.0.1:40028 - "POST /moneymiltonproject/1?api_key=BGHPrLuqySlNDKQ3M76W&name=YOUR_IMAGE&confidence=40&stroke=1&labels=false&format=json HTTP/1.1" 500 Internal Server Error
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `('
-bash: INFO:: command not found
```

Roboflow message

Next Step



EXPORT TO TENSORFLOW
LITE



CREATE AN APP ANDROID
STUDIO



DEPLOY THE MODEL ON
ANDROID STUDIO VIRTUAL
MACHINE

Recap:



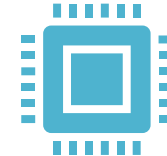
Used Transferring
Learning to get the
model



Train the model using
dataset



Validate the model on
the PC



“Technically” Deployed
a model Raspberry pi
using pi camera